ANATOMY OF AI AGENTS 2026

www.arianprabowo.com · 7-Layer OSI-Style Stack Reference (Based on Roitman arXiv:2606.24937)

What actually runs under the hood of an autonomous AI agent in 2026?
Like the 7-layer OSI model, modern agents form a left-to-right stack — from GPU silicon (L1) &
uv/PyTorch/JAX (L2) to Data/RAG (L3), Arch/RLVR (L4), MCP (L5), Harness^2 (L6) & A2A swarms (L7).

① TRACE LEFT → RIGHT: Follow Layers 1 → 7 from silicon to swarms
② BUILD EACH LAYER: Top 3 pills = flagship 2026 primitives & tools
③ HARDEN PROD: Audit the red FAILURE MODES & TRAPS row at bottom

*Synthesized from Haggai Roitman's 'The Hitchhiker's Guide to Agentic AI: From Foundations to Systems' (arXiv:2606.24937, v1.4, June 2026) + 2026 systems tooling. Companion to AI Skill Tree 2026 & AI Ladder Evolution by Arian Prabowo · Updated: 2026-11-07 23:31:22 AEDT (UTC+11:00)

OSI AGENT STACK

LAYER 1 → LAYER 7

Left-to-Right Pipeline
× 4 Architectural Rows

PILLAR 1

CORE PRIMITIVES & ARCHITECTURE

Hardware Physics, Mathematical Ops, Data Structures & Control Topologies

- Layer 1–2: Silicon/Kernels
- Layer 3–4: Data/Model Math
- Layer 5–7: Memory/Loops/MAS

PILLAR 2

FRAMEWORKS, PKGS & PROTOCOLS

CLI Package Managers, DL/Serving Frameworks, Model Checkpoints & Agent SDKs/Protocols

- brew · uv · conda · pip · venv
- PyTorch · JAX · vLLM · TRL
- MCP · SKILL.md · LangGraph · A2A

PILLAR 3

SCALING, EVALS & SECURITY

Throughput SLAs, MicroVM Sandboxes, Verification Oracles & Zero-Trust Guardrails

- MFU · E2B/bwrap · RAG Triad
- RLVR Graders · MCP OAuth 2.1
- SWE-bench · pass^k · KYA

ANTI-PATTERNS

FAILURE MODES & TRAPS

What Breaks in Prod: Bottlenecks, Collapses, Injections & Swarm Echo Chambers

- OOM & CUDA ABI Hell
- Model Collapse & Reward Hack
- Context Rot & $O(N^2)$ Chatter

LAYER 1 → PHYSICAL & SILICON

GPUs/TPUs/NPUs, HBM3e vs. SRAM hierarchy, CUDA/Triton kernels, and NVLink/InfiniBand fabrics.

- OSI L1 Physical · PDU: FLOPs, HBM Bytes & Warps
- Guide: Ch. 1.6, 2.1, 11.10-14, 30.5 (GPUs & Kernels).

GPU / TPU / NPU Tensor Cores
HBM3e vs. On-Chip SRAM Hierarchy
FlashAttention IO-Aware SRAM Tiling

Streaming Multiprocessors (SMs)	Warp Schedulers (32-Thread SIMT)
Arithmetic Intensity (FLOP/Byte)	Roofline Compute vs. Memory Bound
FP16 / BF16 / FP8 / FP4 Formats	Online Softmax Rescaling (Safe Max)
Unified Memory Architecture (UMA)	Wafer-Scale & SRAM-Only LPU Dies

Register File (256KB/SM)	L1 Cache / Shared Memory	L2 Cache (50MB–128MB)
HBM3e Bandwidth (3.3–8 TB/s)	SRAM Bandwidth (~20–30 TB/s)	Transformer Engine FP8/FP4
Stochastic Rounding	Tensor Memory Accelerator (TMA)	Systolic Array Matrix Multiply
Memory Coalescing	Bank Conflict Avoidance	TDP Power & Liquid Cooling

NVIDIA H100 / H200 / Blackwell B200
Google TPU v5p / v6e Trillium / v7
CUDA · OpenAI Triton · Apple Silicon M4

NVIDIA GB200 NVL72 Rack Scale	AMD Instinct MI300X / MI325X
AWS Trainium2 & Inferentia2	Groq LPU & Cerebras WSE-3 ASIC
NVLink 4/5 (900–1,800 GB/s/GPU)	InfiniBand NDR/XDR (400–800Gb/s)
cuBLAS / cuDNN / CUTLASS / FlashInfer	AMD ROCm / HIP & Apple Metal

DGX / HGX 8-GPU Node Topology	GPUDirect RDMA / Storage	Mellanox ConnectX-7 / 8 NICs
80GB / 141GB / 192GB VRAM Tiers	Mac Mini M4 64GB-512GB Cluster	RTX 4090 / 5090 Local Workstation
On-Device NPU (45-80+ TOPS)	Co-Packaged Silicon Photonics	Vera Rubin NVL144 (HBM4 Roadmap)
Liger-Kernel Fused Triton Ops	ECC Memory Error Correction	NCCL / Gloo Collective Backend

Model FLOPs Utilization (MFU > 45%)
Prefill (Compute) vs. Decode (BW)
Confidential Compute GPU TEEs

Prefill Phase (Matrix-Matrix Gemm)	Decode Phase (Matrix-Vector Gemv)
Multi-Instance GPU (MIG) Slices	Rail-Optimized Fat-Tree Topology
NVIDIA DCGM Hardware Health Check	Silent Data Corruption (SDC) Scans
Nsight Compute & Systems Profiling	Hardware Root-of-Trust Attestation

MFU = Achieved / Peak FLOPs	Hardware FLOPs Utilization (HFU)	Bytes/Param: 2B (BF16) vs 0.5B (FP4)
4-Model RLHF VRAM Multiplier	Optimizer State Memory (12B/param)	Activation Memory Checkpointing
NVLink All-Reduce Bandwidth	Cross-Node All-to-All Bottleneck	Thermal Throttling Mitigation
H100/B200 Secure Enclave (CC-On)	Power Usage Effectiveness (PUE)	Straggler GPU Detection

HBM Memory Bandwidth Wall (Decode)
CUDA Out-of-Memory (OOM) Crashes
Silent Tensor Core Bit-Flip Corrupt

- ▸ Low Decode MFU (<5% Compute Util)
- ▸ PCIe Host-to-Device Bottleneck
- ▸ InfiniBand Link Flaps & Stalls
- ▸ Straggler GPU Halting 1,000 Nodes

4-Model RLHF VRAM Explosion	Thermal Throttling Under RL Rollouts
Unaligned Tensor Shapes (No TC)	Consumer GPU P2P Disabled Locks

LAYER 2 → ENV, PACKAGES, PYTHON & TENSOR

Package managers (`Homebrew`, `uv`, `conda`, `pip`, `venv`), `Python`, `PyTorch`, `JAX`, `vLLM`, and sandboxes.

- OSI L2 Data Link · PDU: Wheels, Tensors & Paged KV
- Guide: Ch. 1.14-15, 11.1-9, 21.5, 22.1-9 (Env & Runtime).

Virtual Environments, Locks & Sandboxes
Autograd Computation Graphs & JIT
PagedAttention KV & Continuous Batch

POSIX PTYs, Processes & Asyncio Loop	ABI / manylinux Wheels & uv.lock
OS Namespaces, cgroups & Seccomp	Dynamic Autograd DAG & torch.compile
PagedAttention Virtual Memory KV	RadixAttention Automatic Prefix Cache
Speculative Decoding (Draft + Verify)	3D Parallelism (DP/TP/PP/SP) & ZeRO

pyproject.toml (PEP 517 / 621)	Content-Addressable Global Cache	Hardlink / Reflink Env Cloning
LD_LIBRARY_PATH / DYLD RPATH	torch.compile (TorchInductor/Dynamo)	JAX jit / vmap / pmap / shard_map
XLA HLO Graph Fusion & CUDA Graphs	Disaggregated Prefill-Decode Routing	Chunked Prefill & Guided Decoding
KV-Cache FP8 / INT4 Quantization	Container UnionFS / OverlayFS Layers	Hardware Virtualization (KVM / HVF)

Homebrew (brew) · uv · conda · pip · venv
Python 3.12+ · PyTorch 2.x · JAX · MLX
vLLM · SGLang · llama.cpp · Docker · E2B

Homebrew (brew) & System Toolchains	uv (uv venv / uv run / uvx / uv pip)
conda / Miniforge / mamba / Pixi / Poetry	pip · venv · virtualenv · pipx
PyTorch 2.x · JAX / Flax · Apple MLX	vLLM · SGLang · TensorRT-LLM · Ollama
Docker / OCI & NVIDIA Container Toolkit	Rust (PyO3/Maturin) & TypeScript/Bun

brew install uv git ripgrep fd tmux	uv lock & uv sync --frozen	environment.yml & conda-lock
pip install --no-build-isolation	DeepSpeed ZeRO & PyTorch FSDP2	llama.cpp (GGUF) & ExLlamaV2/v3
Outlines & XGrammar (JSON/CFG)	Ray Serve & NVIDIA Triton Server	ONNX Runtime & Safetensors Zero-Copy
Nix / Devcontainers / Direnv	Slurm / Kubernetes (K8s) / Ray	Pydantic v2 (Rust Schema Core)

E2B / Firecracker MicroVM Sandboxes
OS Sandboxing (Bubblewrap / Seatbelt)
TTFT / TPOT SLAs & Async RL Sync

gVisor User-Space Kernel Isolation	macOS sandbox-exec & Linux bwrap
OpenEnv Standardized Gym Reset/Step	Air-Gapped VLAN & Egress Firewall
Scoped Ephemeral OAuth / API Tokens	NCCL Weight Broadcast (Trainer->vLLM)
Per-Tenant LoRA Adapter Hot-Swapping	Grammar-Constrained 100% Valid JSON

env.reset() / env.step(action) API	SWE-bench Docker Repo Images	OSWorld / WebArena Live Desktop VMs
TTFT < 200ms · TPOT < 15ms/tok	Goodput (SLO-Compliant Tok/s)	In-Flight Weight Update (Zero Stall)
S-LoRA / Punica Multi-Adapter Batch	Pyodide / Wasm Code Execution	Secret Scrubbing in Env Vars
SBOM & pip-audit Vulnerability Scan	Signed Container Attestation (Cosign)	Deterministic Seed Mode (cuDNN)

Python Dependency & CUDA ABI Hell
Unsandboxed Host RCE & rm -rf /
KV-Cache Fragmentation & Preemption

- ▸ nvcc vs. torch.version.cuda Mismatch
- ▸ Global System Python Pollution
- ▸ pip install Breaking flash-attn ABI
- ▸ Typosquatted PyPI / npm / Skill Packs

Agent Leaking .env & SSH Keys	60-80% Contiguous KV Memory Waste
RL Rollout Generation Bottleneck	torch.compile Graph-Break Thrashing

LAYER 3 → DATA, TOKENIZATION & RAG

Pre-training/SFT/RLVR data curation, synthetic trajectories, BPE tokenizers, vector DBs, and Agentic RAG.

- OSI L3 Network · PDU: Token IDs, Embeddings & Chunks
- Guide: Ch. 1.2, 3.2-5, 4.3, 16, 18.4, 30.2 (Data & RAG).

BPE Tokenization, Packing & Loss Masking
SFT, Preference & RLVR Data Curation
Embeddings, Vector HNSW & Hybrid RAG

Byte-Level BPE / SentencePiece / Tiktoken	Sequence Packing & Position-ID Reset
Completion-Only Loss Masking (Labels=-100)	Chosen vs. Rejected Preference Pairs
Synthetic Reasoning & Trajectory Gen	Bi-Encoder (HNSW) vs. Cross-Encoder
Hybrid RAG (Dense + BM25 + RRF Fusion)	GraphRAG Entity Summaries & REFRAG

MinHash LSH & Exact Substring Dedup	FastText / KenLM Quality Filtering	Decontamination (N-gram / Embed Match)
ChatML Special Tokens (<|im_start|>)	Rejection Sampling (Best-of-N Filter)	Execution-Verified Code/Math Traces
Contextual Chunk Headers (Anthropic)	Late Interaction ColBERT Token MaxSim	Search-R1 RL Query Reformulation
Self-RAG Critique Tokens (ISREL/ISSUP)	Corrective RAG (CRAG) Web Fallback	Matryoshka Representation (MRL)

HF Datasets · Datatrove · NeMo Curator
Tiktoken · SentencePiece · HF Tokenizers
Qdrant · Milvus · pgvector · Neo4j

FineWeb-Edu · Dolma · The Stack v2	Apache Arrow · Parquet · WebDataset
Docling · Unstructured · Marker PDF	Tree-sitter AST Codebase Chunking
BGE-M3 · Cohere Embed v3 & Rerank v3	FAISS · LanceDB · Chroma · Weaviate
LlamaIndex · Haystack · LightRAG	Distilabel · Curator Synthetic Suites

Polars · DuckDB · PyArrow ETL	Ray Data Distributed Streaming	MosaicML StreamingDataset (MDS)
OpenMathReasoning & NuminaMath	SWE-Gym & SWE-smith Trajectories	ShareGPT & OpenAI Harmony Formats
Firecrawl · Jina Reader Markdown	Graphiti & Cognee Knowledge Graphs	SQLite-vec & DuckDB-VSS Local
DVC & LakeFS Data Version Control	Cleanlab Automated Label Audit	Argilla Human-in-the-Loop Curation

RAG Triad Evals (Faithfulness / Recall)
Benchmark Decontamination & PII Scrub
Document-Level RBAC & Injection Scans

Ragas & TruLens Groundedness Oracles	NDCG@k · MRR · Recall@k Retrieval Audits
Pre-Indexing ACL / RBAC Permission Tags	Regex + Presidio PII & Secret Scrubbing
Canary String & N-Gram Contamination	Synthetic Data Diversity & Entropy Audit
Citation Span-Locking & Provenance	Indirect Prompt Injection Sanitization

LLM-as-a-Judge Data Quality Scoring	Perplexity Outlier Filtering (KenLM)	Topic & Difficulty Curriculum Mix
Needle-in-a-Haystack (NIAH) & RULER	FRAMES & LongBench Multi-Hop Eval	Hallucination Span (LettuceDetect)
Poisoned Document Backdoor Scans	Copyright & SPDX License Filtering	Cross-Tenant Vector Namespace Auth
Embedding Drift & Stale Index Check	Tokenizer Fertility (Tok/Word) Audit	Zero-Retention Enterprise Pipelines

Garbage-In Synthetic Model Collapse
Naive Fixed-Size Mid-Table Chunking
Indirect Prompt Injection via RAG

- ▸ Training on Prompt Tokens (No -100 Mask)
- ▸ Test-Set Contamination Mirage
- ▸ Cross-Row Table & Function Splitting
- ▸ Unfiltered Vector RAG Leaking Docs

Chat Template EOS/BOS Token Mismatch	Tokenizer Glitch & Byte-Fallback Bloat
Unverified Synthetic Hallucinations	Hallucinated Citations on Unread Text

LAYER 4 → MODEL ARCHITECTURE & RLVR

Transformer/MoE/hybrid architectures, SFT, PEFT adapters (`LoRA`), preference tuning (`DPO`), and `RLVR` (`GRPO`).

- OSI L4 Transport · PDU: Logits, Adapters & CoT
- Guide: Ch. 1, 2.2, 3-10, 12-13, 30.1-3 (Models & RL).

Transformer · Sparse MoE · GQA / MLA
Hybrid Linear Attention (Mamba/DeltaNet)
PEFT (LoRA/DoRA) & RLVR (PPO/DPO/GRPO)

Pre-Norm RMSNorm · SwiGLU · RoPE/YaRN	Grouped-Query (GQA) & Latent (MLA)
Sparse MoE Top-K & Shared Experts	Gated DeltaNet / Mamba-2 SSM Hybrid
LoRA · QLoRA (NF4) · DoRA · PiSSA	Preference Tuning: PPO · DPO · KTO
Critic-Free RLVR: GRPO · Dr.GRPO · DAPO	Harness Distillation (Harness-Zero)

Multi-Head Latent Attention (MLA)	Aux-Loss-Free MoE Load Balancing	rsLoRA (alpha/sqrt(r)) · QLoRA NF4
Zeroth-Order MeZO Forward-Only Tune	Bradley-Terry RM & Process RM (PRM)	IPO · ORPO · SimPO · Online DPO
RLCR Calibrated Brier Score Reward	On-Policy Self-Distillation (OPSD/RLCSD)	Test-Time Compute (Best-of-N / MCTS)
Cold-Start Pure-RL Emergence (R1-Zero)	MergeKit (SLERP / TIES / DARE)	AdamW vs. Muon Matrix Orthogonal

GPT-5.4 / o3 · Claude 4.6/5 · Gemini 3
DeepSeek-R1/V3.2 · Qwen 3 · Llama 4
HF TRL · PEFT · verl · OpenRLHF · Unsloth

TRL (SFTTrainer / DPOTrainer / GRPO)	verl (HybridEngine RLHF/RLVR at Scale)
OpenRLHF (Ray + vLLM Distributed RL)	Unsloth (2x Speed Triton Backprop)
Axolotl & LLaMA-Factory YAML Suites	HuggingFace Transformers & PEFT
Prime Intellect verifiers & OpenR1	Torchtune & NeMo Aligner

DeepSeek-R1-Distill (7B-70B Weights)	Qwen3-235B-A22B & QwQ-32B	Claude Sonnet 4.5 / Opus 4.6 / 5.5
OpenAI GPT-5.4-Codex & o4-mini	Gemini 2.5 / 3.1 Pro & 3.8 Flash	Mistral Large 3 & Codestral
bitsandbytes (NF4) · AWQ · GPTQ	Lean 4 / SymPy Formal Math Verifiers	MergeKit CLI Checkpoint Merger
WandB / TensorBoard / MLflow Logs	vLLM Rollout Worker Integration	BitNet.cpp 1.58-Bit Ternary Weights

Verifiable Execution Oracles (RLVR)
KL-Divergence & Entropy Regularization
Constitutional AI & Safety Classifiers

Deterministic Unit-Test / Math Graders	DAPO Clip-Higher & Dynamic Sampling
Multi-Objective Reward Combination	Listwise Rank-Based Reward Models
Instruction Hierarchy Privilege Training	Deliberative Alignment in CoT Traces
Activation Steering & SAE Circuit Probes	GCG / AutoDAN / Best-of-N Red-Teaming

KL Penalty beta * D_KL(pi_theta || pi_ref)	PPO Clipping epsilon in [0.1, 0.28]	GRPO Group Size G = 8..64 Rollouts
Pass@k vs. Majority@k Scaling	Length-Controlled Win-Rate Audits	Catastrophic Forgetting Evals
Sleeper Agent Backdoor Detection	Sandbagging & Alignment Faking Audits	Machine Unlearning (RMU / Grad Ascent)
CBRN / Cyber ASL-3/4 Threshold Gates	Contamination-Free LiveBench / HLE	Calibration Expected Error (ECE)

Reward Hacking & Length Inflation
Entropy Collapse & Mode Collapse
Catastrophic Forgetting / Alignment Tax

- ▸ GRPO Zero-Variance Advantage (All 0/1)
- ▸ DPO Likelihood Displacement (Both Drop)
- ▸ MoE Routing Collapse (Expert Starve)
- ▸ Naive INT4 Quant Breaking Reasoning

Sycophancy & Delusional Validation	Emergent Misalignment from Narrow SFT
Over-Refusal on Benign Triggers	FP16 Loss-Scaling Underflow / NaNs

LAYER 5 → CONTEXT, MEMORY, TOOLS & MCP

Context budgeting, 4-tier agentic memory (`Sleep-Time Compute`), function calling, `MCP`, and `SKILL.md`.

- OSI L5 Session · PDU: Context Windows & JSON-RPC
- Guide: Ch. 17, 18.2-3, 23, 24 (Memory, Tools, MCP & Skills).

Context Engineering & Token Budgeting
4-Tier Memory + Sleep-Time Dreaming
Tool Calling, MCP Bus & Progressive Skills

Working Memory (Active Context Window)	Episodic Memory (Past Experience Logs)
Semantic Memory (Entity/Fact Triples)	Procedural Memory (SKILL.md / Weights)
Sleep-Time Offline Consolidation (Dream)	JSON-RPC 2.0 Host-Client-Server Bus
Progressive Disclosure Skill Loading	Observation Masking & /compact Hygiene

System / History / RAG / Tool Scratchpad	Write / Read / Reflect / Consolidate Ops	Ebbinghaus Temporal Memory Decay
Proactive Context Pre-Fetching	KV-Cache Prefix Pinning (90% Discount)	Parallel Tool-Call Batch Execution
Strict JSON Schema Tool Definitions	Code-Act (Python Execution as Tool)	Dynamic Tool Search (Top-K Schemas)
Rolling Summary & Turn Truncation	Subagent Forked Context Isolation	Scratchpad & Todo.md State Offload

Model Context Protocol (MCP / .mcp.json)
Agent Skills (SKILL.md Folder Standard)
Letta (MemGPT) · Mem0 · Zep · LangMem

MCP Primitives: Tools, Resources, Prompts	MCP Transports: stdio & Streamable HTTP
FastMCP Python / TypeScript SDKs	SKILL.md YAML Frontmatter + Instructions
skills/scripts/ & skills/references/ Dirs	Letta (MemGPT) OS-Paged Memory Blocks
Mem0 · Zep · Graphiti Temporal Memory	Linux Foundation AAIF MCP Registry

MCP Sampling & Roots Client Callbacks	MCP Elicitation User Confirmation	.claude/skills/ & .agents/skills/ Paths
agentskills.io Open Specification	LangMem Semantic/Episodic Store	Cognee & Memory Memory Engines
Redis / Postgres Agent Session Stores	OpenAPI-to-MCP Auto-Generators	Browserbase / Playwright MCP Servers
GitHub / Slack / Postgres MCP Servers	Anthropic & Cursor MCP Marketplaces	Smithery & Glama MCP Registries

MCP OAuth 2.1 Scopes & Tool Signing
/compact Hygiene (<65% Context Cap)
Memory Contradiction & Decay Audits

Human-in-the-Loop Destructive Tool Gate	Tool Schema Token-Bloat Minimization
Memory Contradiction & Stale-Fact Purge	LOCOMO & LongMemEval Memory Evals
MCP Server Code-Signing Verification	Least-Privilege Read-Only Connectors
OX Security stdio RCE Hardening	Cross-Tenant Memory Isolation

Context Utilization < 65% Rot Threshold	Lost-in-the-Middle U-Curve Placement	BFCL (Berkeley Function-Calling) Eval
ToolBench & API-Bank Multi-Tool Eval	Episodic Replay Buffer Deduplication	Tool Output Truncation & Pagination
Skill Static Analysis Before Install	Egress Domain Whitelisting per Tool	OAuth 2.1 PKCE Token Refresh Flow
Idempotency Keys for Mutating Tools	Audit Logging of All JSON-RPC Calls	Canary Tokens in Memory Stores

Context Rot & Lost-in-the-Middle
MCP Tool Bloat (100+ Schemas)
Malicious Skills & MCP Rug-Pulls

- ▸ 100+ MCP Tool Schemas Blowing Context
- ▸ ClawHavoc Malicious SKILL.md Payloads
- ▸ MCP Rug-Pull & Tool Shadowing Attacks
- ▸ Memory Poisoning Across Sessions

Stale Episodic Memory Overriding Code	stdio MCP Server Spawning Host Shells
Malformed Tool-Call JSON Loops	Unbounded Tool Output Flooding Window

LAYER 6 → AGENT HARNESS, LOOP & GRAPH

The CPU/OS of an agent: `CLAUDE.md`/`AGENTS.md`, Loop Engineering, Graph Engineering (`LangGraph`), and `Harness²`.

- OSI L6 Presentation · PDU: State Transitions & Diffs
- Guide: Ch. 18, 19, 20, 21, 27 (Harness, Loops & Graphs).

Agent Harness (Prompt+Tools+State+Loop)
Loop Engineering (Observe→Decide→Act)
Graph Engineering (Typed StateGraphs)

ReAct · Reflexion · Plan-and-Execute	5 Loop Primitives (State/Step/Check/Stop)
Typed State Schema & Reducer Functions	Conditional Edges & Cyclic Graph Routing
Fan-Out / Fan-In Parallel Send() API	Recursive Agent Harnessing (Harness^2)
Spec-Driven Development (Spec→Plan→Task)	Cognitive Escalation & Uncertainty Handoff

pi(a_t | s_t) Inference-Time Policy Loop	LATS (Language Agent Tree Search)	LLM Compiler Parallel DAG Execution
BDI (Belief-Desire-Intention) Hybrid	Checkpointer Time-Travel & Replay	Human-in-the-Loop Interrupt() Gates
Stagnation & Repeat-Action Detection	Subagent Context Isolation (Forked Call)	Red-Green-Refactor TDD Agent Cycle
Exponential Backoff & Schema Self-Repair	Token/Cost/Iteration Termination Caps	Dynamic Model Routing (Fast vs. Reasoner)

Claude Code CLI · Codex CLI · OpenCode
Cursor · Windsurf · Cline / Roo · Aider
LangGraph · OpenAI Agents SDK · DSPy

CLAUDE.md & AGENTS.md Repo Constitutions	.cursor/rules & .github/copilot-instructions
GitHub Spec Kit & obra/superpowers TDD	LlamaIndex Workflows (Event-Driven)
Smolagents (CodeAct Python-as-Action)	PydanticAI Type-Safe Agent Harness
OpenHands (OpenDevin) & Block goose	Semantic Kernel & Haystack Agents

LangGraph PostgresCheckpointer	DSPy MIPROv2 Prompt/Weight Compiler	Claude Agent SDK & Computer Use API
Browser-Use & Playwright DOM Harnesses	Vercel AI SDK Core (generateText/maxSteps)	Guardrails AI & NeMo Guardrails
Instructor & BAML Structured Outputs	Litellm Unified Model Router	Git Pre-Commit & CI Harness Hooks
Repomix / Gitingest Codebase Packers	LangSmith · Arize Phoenix · Langfuse	OpenTelemetry (OTel) GenAI Spans

Decoupled Verification Oracles (No Self-Grade)
SWE-bench Verified/Pro & Terminal-Bench
pass^k Reliability & Trajectory Tracing

Property-Based Testing (Hypothesis/Kani)	Mutation Testing (Stryker / mutmut) Gates
Immutable CI Tests Locked From Agent Write	Trajectory Tool-Call Exact/Prefix Match
LLM-as-a-Judge Pairwise Rubric Grading	Cost-per-Resolved-Task Efficiency Metric
PyRIT & Garak Adversarial Red-Teaming	LlamaFirewall Prompt/Tool Guardrails

pass^k = (pass@1)^k Consistency Metric	METR 50%-Reliability Horizon (2h-30h)	GAIA · OSWorld · WebArena · TAU-bench
Scale SWE-bench Pro (Private Repos)	Step Efficiency (Optimal / Actual Steps)	Deterministic Mock-Tool Replay Harness
Shadow-Mode Traffic Counterfactual Eval	Circuit-Breaker Spend & Loop-Depth Caps	Canary Token Exfiltration Tripwires
Git Diff AST Security Linter (Semgrep)	Automated Rollback on Regression	EU AI Act / SOC2 Immutable Audit Logs

Agent Deleting / Mocking Failing Tests
Unbounded Loop Burn & Action Stagnation
God-Node Anti-Pattern & State Bloat

- ▸ Coupled Agent Rewriting Its Own Grader
- ▸ 0.95^20 = 35.8% Compounding Step Decay
- ▸ Poisoned .cursor/rules or GitHub Issues
- ▸ CVE-2025-53773 IDE Config Auto-Approve RCE

Swallowing Exceptions in Tool Wrappers	Appending Raw 50KB HTML to Graph State
Premature Termination Before Verification	Sycophantic Plan Pivot on User Pushback

LAYER 7 ★ MULTI-AGENT MESH, A2A & UI

Multi-agent topologies (`OWC`), inter-agent protocols (`A2A`/`ACP`), parallel worktree swarms, and Agentic/Generative UI.

- OSI L7 Application · PDU: A2A Tasks & GenUI Streams
- Guide: Ch. 21.6, 25, 26, 28, 30.10/18 (A2A, MAS & Agentic UI).

Orchestrator-Worker-Critic (OWC) Pattern
Multi-Agent Topologies (Hierarchical/Mesh)
Generative UI & Shared State Sync

Centralized Supervisor vs. Decentralized	Hierarchical Tree (Squads of Squads)
Debate / Society-of-Mind Consensus	Blackboard Shared-Memory Coordination
Contract-Net Task Bidding & Auctions	Parallel Git-Worktree Swarm Isolation
A2A (Horizontal) × MCP (Vertical) Dual Bus	Human-on-the-Loop Escalation & Approval

Agent Card (/.well-known/agent.json)	Task Lifecycle (Submitted→Working→Done)	Multi-Agent MDP & CTDE MARL Training
Credit Assignment (Shapley / Counterfactual)	Role Specialization & Persona Diversity	Byzantine Fault-Tolerant Critic Voting
Pub/Sub Asynchronous Event Choreography	Process-Centric Enterprise State Workflows	Server-Driven UI Component Streaming
Optimistic UI Updates & Diff Previews	Ambient OS Cross-App Delegation	Organizational Digital-Twin Simulation

Google A2A Protocol · IBM ACP · ANP
CrewAI · AutoGen / AG2 · LangGraph Swarm
CopilotKit · AG-UI · Vercel AI SDK UI

Google Agent-to-Agent (A2A) JSON-RPC/SSE	IBM BeeAI & Agent Communication Protocol
Microsoft AutoGen v0.4 / Semantic Kernel	CrewAI Flows & Role-Based Crews
Claude Code Subagents & Worktree Swarms	OpenClaw 24/7 Multi-Channel Agent Daemon
Agent Gateways (Kong / Solo / LiteLLM)	Claude Artifacts · ChatGPT/Gemini Canvas

AG-UI (Agent-User Interaction Protocol)	Vercel AI SDK useChat / streamUI Hooks	Chainlit · Streamlit · Gradio Agent UIs
OpenAI ChatKit & Realtime WebRTC Voice	LangGraph Platform Cloud & Studio UI	Camel-AI Workforce & MetaGPT SOPs
OpenAI Swarm / Agents SDK Handoffs	A2A Artifact & Push-Notification Webhooks	React Server Components (RSC) GenUI
Slack / Discord / WhatsApp Bot Bridges	Linear / GitHub PR Async Agent Teammates	1-Click MCP & A2A Enterprise Registries

Agent Gateway (mTLS, Auth, FinOps Caps)
Cross-Agent Trust & Cryptographic KYA
Independent Critic & Red-vs-Blue Swarms

Zero-Trust Agent Identity (SPIFFE / KYA)	OAuth 2.1 On-Behalf-Of (OBO) Token Flow
Per-Swarm Token & Dollar Circuit Breakers	Anti-Collusion & Decorrelated Critic Models
Cross-Repo API Contract Locks (OpenAPI)	Deadlock, Livelock & Cycle Detection
Provenance Watermarking (C2PA / SynthID)	Strict Liability & Action Audit Ledgers

Communication Overhead O(N^2) Pruning	Multi-Agent Judge Independence Check	Cross-Org Data-Loss Prevention (DLP)
Swarm Amplification Factor Monitoring	Canary Deployment of Agent Topologies	Chaos Engineering (Injecting Rogue Agents)
XBOW / Cybench / T3MP3ST Cyber Ranges	Lean 4 / TLA+ Formal Protocol Verification	Human Escalation SLA Latency Tracking
Anti-Sycophancy & Anti-Psychosis Filters	GDPR / HIPAA Cross-Agent Consent Scopes	Kill-Switch Broadcast Across Swarm Nodes

Cascading Hallucination & Swarm Echo
$O(N^2)$ Chatty Agent Token Explosion
Cross-Agent Confused Deputy & Escalation

- ▸ 7-Persona Roleplay Chatter Burning 15x Cost
- ▸ Homogeneous Critic Blindly Agreeing
- ▸ Supervisor Bottleneck & Context Overflow
- ▸ Merge Conflicts Across Parallel Worktrees

Rogue Subagent Bypassing Parent Scopes	Infinite Ping-Pong Handoff Deadlocks
Free-Riding Worker Agents in Crews	UI Stream Desync & Phantom Tool States